

Introduction to Computing: Tutorial 3

BSDS I Year, 2026–2027

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1. Write a Python program to flatten a list that contains nested lists as elements.
 - (a) For simplicity, first assume that each element of the list is itself a simple list containing ‘atomic’ elements. For example, the input list will look like `[[1, 2], [3, 4], [5, 6]]`, and your output should be `[1, 2, 3, 4, 5, 6]`.
 - (b) Extend your program so that it can handle lists whose elements may be either atomic elements or lists containing ‘atomic’ elements. Your input may now look like `[[1, 2], 3, 4, [5, 6]]`. Your output will remain the same as in part (a).
2. Write a program to count the frequency of occurrence of each word in a given piece of text. You should strip apostrophes from the input, so that, for example, `student's` is counted as `student`, and `students'` is counted as `students`. You should print the words in alphabetic order, along with their frequencies.
3. You are given a dictionary d , where the values are all hashable and distinct. Augment the dictionary d with the inverse mapping, i.e., if k is a key in the original dictionary with value v , the augmented dictionary should contain both $\langle k, v \rangle$ and $\langle v, k \rangle$ as items.
4. Recall that the `|=` operator works like a ‘union update’ operator on dictionaries. Write a program that computes the *difference* between two dictionaries in the same spirit. Thus, if your program is given dictionaries `d1` and `d2`, your program should print a dictionary `d3` that contains items corresponding to only those keys from `d1` that do not also occur in `d2`.
5. Write a program that does the following **without** using the Python `set` and `frozenset` types and associated operations.
 - (a) Given a list L of integers, return another list containing only the distinct integers in L .
 - (b) Given two lists L_1 and L_2 of integers, return a list corresponding to the (i) union, and (ii) intersection of the set of integers contained in the two lists.